AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 3: Vector Norms; Matrix Norms

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Outline

1. **Vector Norms (NLA §3)**
2. Matrix Norms (NLA §3)

Definition of Norms

- A norm captures the "size" of a vector or the "distance" between vectors.
- There are many different measures for "sizes", but a norm must satisfy some requirements.

Definition

A *norm* is a function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ that assigns a real-valued length to each vector. It must satisfy the following conditions:

$$
\begin{aligned}
&(1) \quad \|x\| \geq 0, \text{ and } \|x\| = 0 \text{ only if } x = 0, \\
&(2) \quad \|x + y\| \leq \|x\| + \|y\|, \\
&(3) \quad \|\alpha x\| = |\alpha| \|x\|.
\end{aligned}
$$

- An example is the Euclidean length, defined as $\|x\| = \sqrt{\sum_{i=1}^{n} |x_i|^2}$.

p-norms

- Euclidean length is a special case of p-norms, defined as

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$$

  for $1 \leq p \leq \infty$.
- The Euclidean norm is the 2-norm $\|x\|_2$ (i.e., $p = 2$).
- 1-norm: $\|x\|_1 = \sum_{i=1}^{n} |x_i|$.
- ∞-norm: $\|x\|_\infty$. What is its value?

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- 1-norm: $\|x\|_1 = \sum_{i=1}^{n} |x_i|$.
- ∞-norm: $\|x\|_\infty$. What is its value?
  - Answer: $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$.
- Why do we require $p \geq 1$? What happens if $0 \leq p < 1$?

Some Properties of Vector Norms

- Hölder inequality:

$$|x^\top y| \le \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

- In particular, when $p = q = 2$, we have the Cauchy–Schwarz inequality:

$$|x^\top y| \le \|x\|_2 \|y\|_2.$$

- All norms are *equivalent* in that there exist constants c_1 and c_2 such that

$$c_1 \|x\|_\alpha \le \|x\|_\beta \le c_2 \|x\|_\alpha,$$

  where c_1 and c_2 may depend on n. For example,

$$\|x\|_2 \le \|x\|_1 \le \sqrt{n} \|x\|_2.$$

- The 2-norm is preserved under orthogonal transformation Qx:

$$\|Qx\|_2 = \|x\|_2.$$

Weighted p-norms

- A generalization of the p-norm is the *weighted p-norm*, which assigns different weights (priorities) to different components.
  - It is anisotropic instead of isotropic.
- Algebraically, $\|x\|_W = \|Wx\|_p$, where W is a diagonal matrix with the ith diagonal entry $w_i \neq 0$ being the weight for the ith component.
- In other words,

$$\|x\|_W = \left(\sum_{i=1}^{n} |w_i x_i|^p\right)^{1/p}.$$

- What happens if we allow $w_i = 0$?

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- What happens if we allow $w_i = 0$?
- Can we further generalize it to allow W to be an arbitrary matrix?
  - No. But we can allow W to be an arbitrary nonsingular matrix.

A-norm of Vectors

- Given a positive definite matrix $A \in \mathbb{R}^{n \times n}$, the A-norm on $\mathbb{R}^n$ is

$$\|x\|_A = \sqrt{x^\top A x}.$$

- Note: The weighted 2-norm with W corresponds to the A-norm with $A = W^2$.
- These conventions are somewhat inconsistent, but they are both commonly used in the literature.

Outline

1. Vector Norms (NLA §3)
2. **Matrix Norms (NLA §3)**

Frobenius Norm

- One can define a norm by viewing an $m \times n$ matrix as a vector in $\mathbb{R}^{mn}$.
- One useful norm is the Frobenius (or Hilbert–Schmidt) norm:

$$\|A\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{m}|a_{ij}|^2} = \sqrt{\sum_{j=1}^{n}\|a_j\|_2^2}.$$

  This is the 2-norm of the (mn)-vector.
- Furthermore,

$$\|A\|_F = \sqrt{\mathrm{tr}(A^\top A)},$$

  where $\mathrm{tr}(B)$ (the trace of B) is the sum of its diagonal entries.
- Note that for $A \in \mathbb{R}^{n\times\ell}$ and $B \in \mathbb{R}^{\ell\times m}$,

$$\|AB\|_F \le \|A\|_F\|B\|_F,$$

  because

$$\|AB\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}|a_i^\top b_j|^2 \le \sum_{i=1}^{n}\sum_{j=1}^{m}\left(\|a_i^\top\|_2\|b_j\|_2\right)^2 = \|A\|_F^2\|B\|_F^2.$$

General Definition of Matrix Norms

- However, viewing an $m \times n$ matrix as a vector in $\mathbb{R}^{mn}$ is not always useful, because matrix operations do not behave this way.
- Similar to vector norms, *general matrix norms* have the following properties (for $A, B \in \mathbb{R}^{m \times n}$):

$$
\begin{aligned}
&(1) \quad \|A\| \geq 0, \text{ and } \|A\| = 0 \text{ only if } A = 0, \\
&(2) \quad \|A + B\| \leq \|A\| + \|B\|, \\
&(3) \quad \|\alpha A\| = |\alpha| \|A\|.
\end{aligned}
$$

- In addition, a matrix norm for $A, B \in \mathbb{R}^{n \times n}$ typically satisfies the *submultiplicativity* property:

$$\|AB\| \leq \|A\| \|B\|,$$

which is a generalization of the Cauchy–Schwarz inequality.

Matrix Norms Induced by Vector Norms

- Matrix norms can be *induced* from vector norms, which can better capture the behaviors of matrix-vector multiplications.

Definition

Given vector norms $\|\cdot\|_{(n)}$ and $\|\cdot\|_{(m)}$ on the domain and range of $A \in \mathbb{R}^{m\times n}$, respectively, the induced matrix norm $\|A\|_{(m,n)}$ is the smallest number $C \in \mathbb{R}$ for which the following inequality holds for all $x \in \mathbb{R}^n$:

$$\|Ax\|_{(m)} \leq C\|x\|_{(n)}.$$

- It is the supremum of $\|Ax\|_{(m)}/\|x\|_{(n)}$ for all $x \in \mathbb{R}^n \setminus \{0\}$.
- It represents the maximum factor by which A can "stretch" $x \in \mathbb{R}^n$:

$$\|A\|_{(m,n)} = \sup_{x\in\mathbb{R}^n, x\neq 0} \frac{\|Ax\|_{(m)}}{\|x\|_{(n)}} = \sup_{x\in\mathbb{R}^n, \|x\|_{(n)}=1} \|Ax\|_{(m)}.$$

- Are vector norms consistent with matrix norms of an $m \times 1$ matrix?

1-norm

- By definition,

$$\|A\|_1 = \sup_{x \in \mathbb{R}^n, \|x\|_1 = 1} \|Ax\|_1.$$

- What is it equal to?

1-norm

- By definition,

$$\|A\|_1 = \sup_{x \in \mathbb{R}^n, \|x\|_1 = 1} \|Ax\|_1.$$

- What is it equal to?
  - Maximum of the 1-norm of the column vectors of A.
  - Or the maximum of the column sums of the absolute values of A ("column-sum norm").
- To show it, note that for $x \in \mathbb{R}^n$ and $\|x\|_1 = 1$:

$$\|Ax\|_1 = \left\| \sum_{j=1}^{n} x_j a_j \right\|_1 \leq \sum_{j=1}^{n} |x_j| \|a_j\|_1 \leq \max_{1 \leq j \leq n} \|a_j\|_1 \|x\|_1.$$

- Let $k = \arg\max_{1 \leq j \leq n} \|a_j\|_1$. Then $\|Ae_k\|_1 = \|a_k\|_1$, so $\max_{1 \leq j \leq n} \|a_j\|_1$ is a tight upper bound.

∞-norm

- By definition,

$$\|A\|_\infty = \sup_{x \in \mathbb{R}^n, \|x\|_\infty = 1} \|Ax\|_\infty.$$

- What is $\|A\|_\infty$ equal to?

∞-norm

- By definition,

$$\|A\|_\infty = \sup_{x \in \mathbb{R}^n, \|x\|_\infty = 1} \|Ax\|_\infty.$$

- What is $\|A\|_\infty$ equal to?
  - Maximum of the 1-norm of the column vectors of $A^\top$.
  - Or the maximum of the row sums of the absolute values of A ("row-sum norm").
- To show it, note that for $x \in \mathbb{R}^n$ and $\|x\|_\infty = 1$:

$$\|Ax\|_\infty = \max_{1 \le i \le m} |a_{i,:} x| \le \max_{1 \le i \le m} \|a_{i,:}^\top\|_1 \|x\|_\infty,$$

  where $a_{i,:}$ denotes the ith row vector of A and $\|a_{i,:}^\top\|_1 = \sum_{j=1}^n |a_{ij}|$.
- Furthermore, $\|a_{i,:}^\top\|_1$ is a tight bound.
- Which vector can we choose for x for equality to hold?

2-norm

- What is the 2-norm of a matrix?

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  Answer: Its largest singular value, which we will explain in later lectures.
- What is the 2-norm of a diagonal matrix D?

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  Answer: Its largest singular value, which we will explain in later lectures.
- What is the 2-norm of a diagonal matrix D?
  Answer: $\|D\|_2 = \max_{i=1}^{n}\{|d_{ii}|\}$.
- What is the 2-norm of a rank-one matrix $uv^\top$? Hint: Use the Cauchy–Schwarz inequality.

2-norm

- What is the 2-norm of a matrix?
  Answer: Its largest singular value, which we will explain in later lectures.
- What is the 2-norm of a diagonal matrix $\boldsymbol{D}$?
  Answer: $\|\boldsymbol{D}\|_2 = \max_{i=1}^{n}\{|d_{ii}|\}$.
- What is the 2-norm of a rank-one matrix $\boldsymbol{u}\boldsymbol{v}^\top$? Hint: Use the Cauchy–Schwarz inequality.
  Answer: $\|\boldsymbol{u}\boldsymbol{v}^\top\|_2 = \|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2$.

Bounding Matrix-Matrix Multiplication

- Let A be an $\ell \times m$ matrix and B an $m \times n$ matrix. Then

$$\|AB\|_{(\ell,n)} \leq \|A\|_{(\ell,m)}\|B\|_{(m,n)}.$$

- To show it, note that for $x \in \mathbb{R}^n$,

$$\|ABx\|_{(\ell)} \leq \|A\|_{(\ell,m)}\|Bx\|_{(m)} \leq \|A\|_{(\ell,m)}\|B\|_{(m,n)}\|x\|_{(n)}.$$

- In general, this inequality is not an equality.
- In particular, $\|A^p\| \leq \|A\|^p$ but $\|A^p\| \neq \|A\|^p$ in general for $p \geq 2$.

Invariance under Orthogonal Transformation

- Given a matrix $Q \in \mathbb{R}^{\ell \times m}$ with $\ell \geq m$. If $Q^\top Q = I$, then Qx for $x \in \mathbb{R}^m$ corresponds to an orthogonal transformation to a coordinate system in $\mathbb{R}^\ell$.
- If $Q \in \mathbb{R}^{m \times m}$, then Q is said to be an *orthogonal* matrix.

Theorem

For any $A \in \mathbb{R}^{m \times n}$ and $Q \in \mathbb{R}^{\ell \times m}$ with $Q^\top Q = I$ and $\ell \geq m$, we have

$$\|QA\|_2 = \|A\|_2 \text{ and } \|QA\|_F = \|A\|_F.$$

In other words, the 2-norm and Frobenius norms are invariant under orthogonal transformation.

Proof for 2-norm: $\|Qy\|_2 = \|y\|_2$ for $y \in \mathbb{R}^m$, and therefore $\|QAx\|_2 = \|Ax\|_2$ for $x \in \mathbb{R}^n$. It then follows from the definition of the 2-norm.

Proof for Frobenius norm: $\|QA\|_F^2 = \operatorname{tr}\left((QA)^\top QA\right) = \operatorname{tr}\left(A^\top Q^\top QA\right) = \operatorname{tr}\left(A^\top A\right) = \|A\|_F^2.$